

COUNTY-LEVEL EVIDENCE BRIEF

LACK OF INSURANCE ACCESS, MORE HOUSING INSECURITY HIGHER DIABETES RATES

Across 3,143 U.S. counties, higher rates of lack of insurance working-age adults and housing insecurity are associated with higher diabetes prevalence.

ANALYSIS SCOPE

U.S. counties

N=3,143

DATA SOURCE

CDC BRFSS data

UNIT OF ANALYSIS

County level

COUNTY-LEVEL ASSOCIATION

As lack of insurance rises, so does diabetes prevalence

READOUT

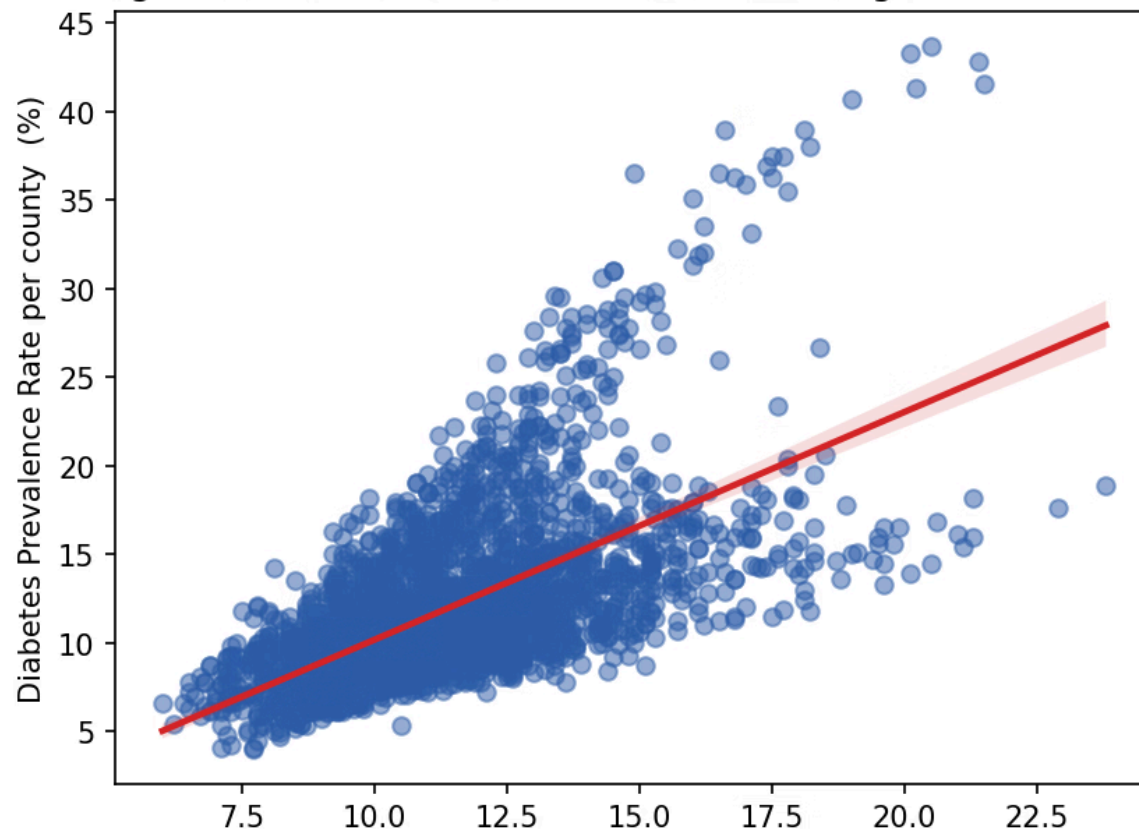
Positive county-level relationship

UNIT

ONE DOT • ONE COUNTY

CDC BRFSS

LACK OF INSURANCE RATE (%) × DIABETES PREVALENCE RATE (%)



POSITIVE LINEAR TREND • lack of insurance ~5-25% • Diabetes Prevalence ~0-45%

RELATIVE EFFECTS

Housing insecurity outweighs insurance access

- Housing insecurity: ~2x effect of lack of insurance rates
- Both factors together: ~74% of variation explained
- Limit: 26% unexplained — income, food insecurity, food access

HOUSING VS. INSURANCE

2x effect

74% explained

26% unexplained

PRACTICE IMPLICATIONS

Implications for health teams

01

Pair support

Insurance enrollment
Housing assistance

02

Prioritize overlap

Overlapping-challenge counties

03

Explore drivers

Income · food insecurity · food
access

INSURANCE ACCESS + HOUSING SECURITY · INTEGRATED COUNTY RESPONSE

Background & methods

01 Research question

Lack of insurance access + housing insecurity predicting diabetes prevalence

02 Data

CDC Behavioral Risk Factor Surveillance System
Phone survey

03 Sample

229,000+ records → **3,143** counties
One row per county

04 Method

Linear Regression Model
Lack of insurance rate + housing insecurity rate → diabetes prevalence

WHY IT MATTERS

Uncontrolled blood sugar

Organ failure

Heart and kidney disease

Blindness